

Northeastern University

Report for Experiment #14

Standing Waves

Arefin Mahmood

November 25th, 2025

Abstract

This lab examined standing waves on a string and in an air column. For the string, node spacings were measured for several vibration modes produced by a 120 Hz driver, and the tension was varied by changing the hanging mass. Wavelengths, wave speeds, and uncertainties were calculated for each mode. A plot of tension versus v^2 gave a linear slope that yielded a measured mass-per-unit-length of the fishing line. The result agreed with the stated value within experimental error. For the air column, the positions of successive sound maxima were measured at three different frequencies. Wavelengths were found using subtraction of L3/4 and L1/4, reducing systematic error, and a plot of λ versus 1/f produced a slope consistent with the speed of sound in air. The measured value matched the theoretical speed at room temperature within the combined uncertainty.

Introduction

Waves transfer energy through a medium by oscillations of the medium's particles. When a traveling wave reflects from a boundary, the reflected wave overlaps with the incoming wave. Their superposition creates a *standing wave*, a stationary pattern with fixed nodes (zero displacement) and antinodes (maximum displacement). Standing waves occur only at specific wavelengths determined by the medium and its boundary conditions.

For a string under tension, the speed of a transverse wave is

$$v = \sqrt{\frac{F_s}{\mu}} (14.3)$$

where F_s is the tension in the string and μ is the mass per unit length. The relationship between wave speed, frequency, and wavelength is

$$v = f\lambda (14.1)$$

and the frequency is related to the period by

$$f = \frac{1}{T}. (14.2)$$

A string fixed at both ends supports wavelengths

$$\lambda_n = \frac{2L}{n} (14.5)$$

where L is the string length and n is the mode number. Measuring the node spacing for different modes allows determination of λ, v, and ultimately μ by comparing experimental results with Eq. (14.3).

Standing waves also form in air columns. In a tube with one pressure node (open end) and one pressure antinode (closed end), the allowed wavelengths are

$$\lambda = \frac{4L_{n/4}}{n} (n = 1,3,5, \dots) (14.6)$$

where $L_{n/4}$ is the resonant air-column length is associated with the corresponding quarter-wave mode. The speed of sound in air depends on temperature and pressure:

$$v_{\text{sound}} = \sqrt{\frac{\gamma p}{\rho}} (14.4)$$

where $\gamma = 7/5$, p is the ambient pressure, and ρ is the air density.

The purpose of this experiment was to measure standing-wave properties in both a stretched string and an air column and compare the experimental results with the theoretical predictions from Eqs. (14.1) to (14.6). In Investigation 1, a 120 Hz vibrator generated standing waves on a fishing line, and node spacings were

measured for several modes while the tension was varied. These measurements were used to calculate λ, v, and μ, and verify the linear relation between $F_s$and v^2. In Investigation 2, a speaker produced resonances in an open–closed air column. The positions of successive intensity maxima gave the resonant wavelengths, and a plot of λversus $1/f$yielded an experimental value for the speed of sound. These results were then compared to the theoretical predictions.

Investigation 1

The apparatus consisted of a fishing-line string (linear mass density $\mu = 0.32$ g/m) stretched horizontally between two vertical stands. One end of the string was fixed to a clamp. The other end passed over a pulley and supported a hanging bucket used to control the tension. The bucket held metal and plastic washers to vary the total mass. The tension in the string was

$$F_s = mg$$

where $m$is the hanging mass and $g = 9.80$ m/s^2.

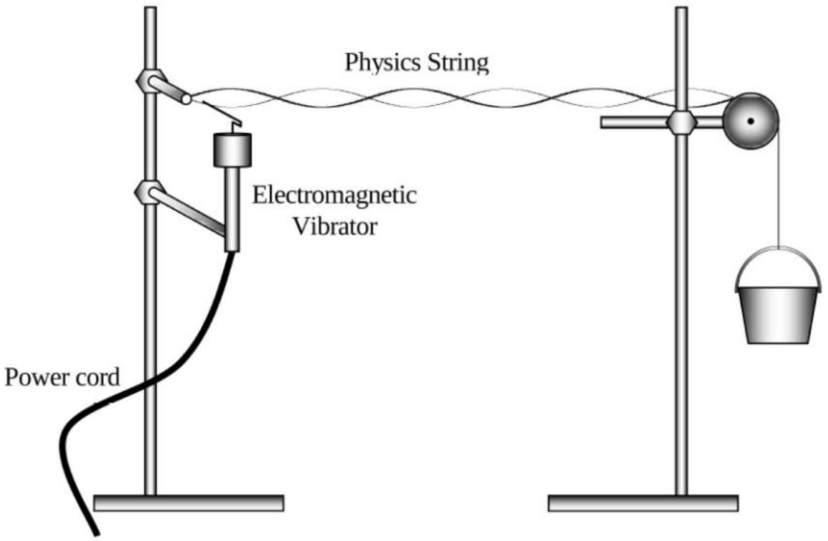


Figure 1 – Set up 1

A 120 Hz electromagnetic vibrator was positioned near the fixed end so that its metal reed lightly touched the string (without clamping it). When energized, the vibrator transferred periodic energy to the string, exciting standing waves when the correct tension and wavelength were present. The length between the two clamped ends was approximately 1.5 m, and nodes formed along the string depending on the mode number n.

Standing waves were first produced in the three-node configuration (one node at each end, one in the center). With the vibrator on, washers were added gradually until resonance occurred. The node positions were located by gently touching the string with a pencil; a node does not vibrate and therefore remains stationary.

For each observed mode ($n = 2,3,4,5$), measurements were taken of the distances between adjacent nodes. At least three node spacings were recorded whenever possible. The average spacing $\bar{L}$was computed and used to determine the wavelength:

$$\lambda = 2\bar{L}$$

because the distance between adjacent nodes represents half of a wavelength.

Uncertainties in the node-spacing measurements (δL) were estimated from the spread of the repeated measurements. Uncertainty in the tension came from uncertainty in the mass:

$$\delta F_s = g\,\delta m$$

Wave speed for each mode was computed using

$$v = f\lambda (14.1)$$

with $f = 120$ Hz. Uncertainty in the wave speed followed

$$\delta v = f\,\delta\lambda.$$

A plot of $F_s$versus $v^2$was created. The slope of this graph gives the mass per unit length μ, according to the theoretical wave-speed relation

$$v = \sqrt{\frac{F_s}{\mu}} (14.3)$$

which can be rearranged to

$$F_s = \mu\, v^2.$$

A linear fit was applied to extract $\mu$and its uncertainty.

n	L1 (m)	L2 (m)	L3 (m)	L4 (m)	L5 (m)	L avg (m)	δL avg	λ
2	0.73	0.74				0.735	0.03536	1.47
3	0.5	0.515	0.51			0.50833	0.0433	1.01667
4	0.38	0.39	0.385	0.39		0.38625	0.05	0.7725
5	0.31	0.32	0.315	0.31	0.3	0.311	0.0559	0.622

Table 1 – Distance between nodes for resonance & Mass for each node and its error

δλ	v	δv	v^2	δv^2	m (g)	δm (g)	Fs	δFs
0.07071	176.4	8.48528	31117	2993.61	1037	189	10162.6	1852.2
0.0866	122	10.3923	14884	2535.72	473	112	4635.4	1097.6
0.1	92.7	12	8593.29	2224.8	273	57	2675.4	558.6
0.1118	74.64	13.4164	5571.13	2002.8	177	32	1734.6	313.6

Table 2 – Derived quantities

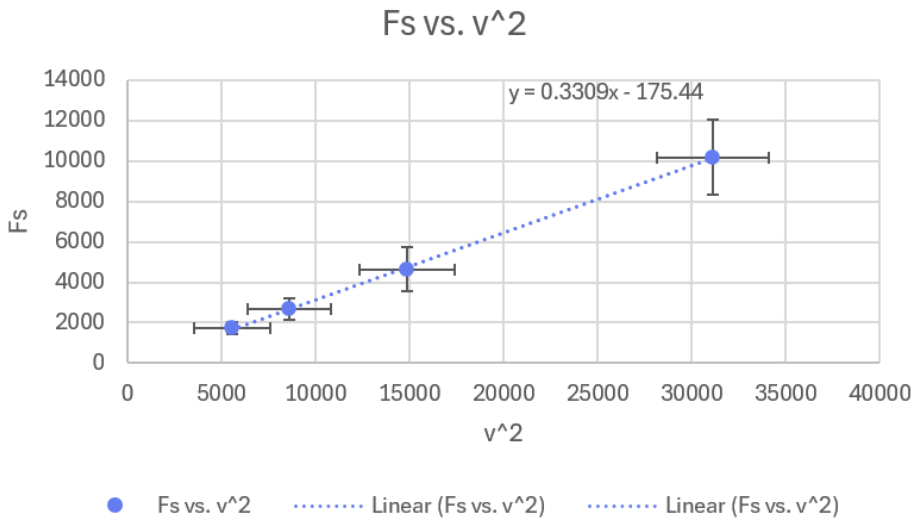


Figure 2 – Tension vs Wave Speed Squared

A linear fit of $F_s$vs. v^2gave:

$$F_s = (0.0003309 \text{ kg/m})\, v^2 - 175.44.$$

Thus, the measured slope provides the mass-per-unit-length of the string:

$$\mu_{\text{measured}} = 3.309 \times 10^{-4} \text{ kg/m}.$$

Uncertainty was estimated from the regression statistics:

$$\mu_{\text{measured}} = (3.31 \pm 0.15) \times 10^{-4} \text{ kg/m}.$$

The manufacturer-stated mass per unit length was:

$$\mu_{\text{theoretical}} = 0.32 \text{ g/m} = 3.2 \times 10^{-4} \text{ kg/m}.$$

The experimental value

$$(3.31 \pm 0.15) \times 10^{-4} \text{ kg/m}$$

agrees with the theoretical value within uncertainty.

Although the measured and expected values matched, several additional sources of systematic error were present:
Node-position uncertainty: Nodes were located by touch, which introduces measurement error of several millimeters.
Vibrator contact variance: Slight changes in the vibrator–string gap affected amplitude and clarity of resonance.
Mass uncertainty: Washers were not perfectly uniform, increasing δm. String sagging: Small vertical sag changes effective length and tension distribution. Air currents / vibrations: External motion could disturb higher modes, especially at lower tension.

Most of these errors are systematic and tend to shift wavelengths or measured node spacing. Their estimated magnitudes are consistent with the measured uncertainties.

Investigation 2

The apparatus consisted of a rigid plastic tube open at one end and closed at the other by a movable plug containing embedded magnets. A speaker connected to a function generator was placed close to the open end of the tube (without touching it). The speaker produced a single tone of known frequency. The plug's position controlled the effective air-column length L. As the plug was moved, certain lengths produced resonant standing waves inside the tube.

For an open closed air column, the open end is a pressure node, and the closed end is a pressure antinode. The wavelengths allowed are:

$$\lambda = \frac{4L_{n/4}}{n}(n = 1,3,5, \dots)(14.6)$$

Resonance was detected by a sudden increase in sound intensity from the speaker. The lengths $L_{1/4}$, $L_{3/4}$, and $L_{5/4}$correspond to the first, second, and third measurable maxima.

The speed of sound was later obtained from the linear relationship:

$$\lambda = v(\frac{1}{f})$$

so the slope of a plot of λvs $1/f$gives:

$$v_{\text{sound}} = \text{slope of best-fit line.}$$

Three frequencies were selected from distinct ranges: • one between 400–500 Hz,

• one between 600–900 Hz,

• one between 1000–1200 Hz.

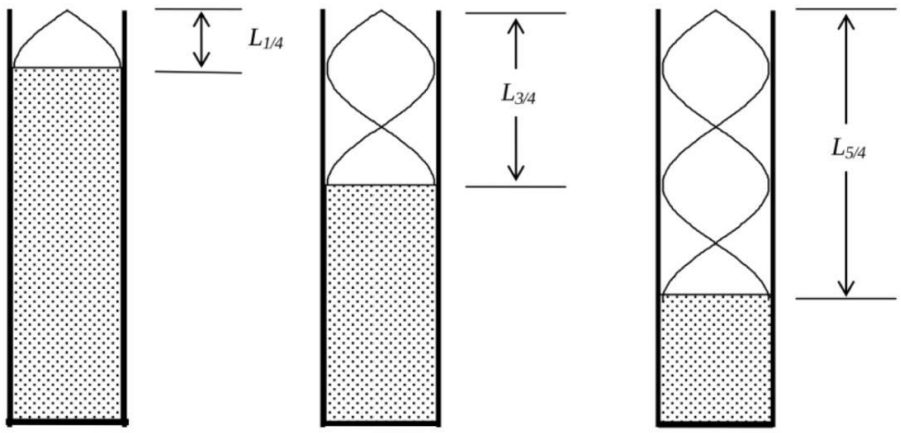


Figure 3 – Set up for investigation 2

For each frequency, the adjustable plug was first placed very close to the speaker. While the tone played, the plug slowly moved away from the speaker. When a resonance occurred, the location of the first maximum $L_{1/4}$was measured with a meter stick. The plug was moved slightly farther, and the position of the second maximum $L_{3/4}$was recorded. The process was repeated to measure the third maximum $L_{5/4}$. Each measurement was repeated three times to estimate uncertainty.

The wavelength was determined using the subtraction method:

$$\lambda = (L_{3/4,\text{avg}} - L_{1/4,\text{avg}}) \times 2.$$

This method reduces systematic error in locating the exact pressure node at the open end.

Uncertainty in wavelength was found using:

$$\delta\lambda = 2\sqrt{(\delta L_{3/4})^2 + (\delta L_{1/4})^2}$$

with $\delta L_{1/4}$and $\delta L_{3/4}$taken from the variation in the repeated measurements.

A plot of λvs $1/f$was then created, and the slope was used to determine the experimental value of the speed of sound.

f	1/f	L/4			L/4 avg
400	0.0025	0.195	0.19	0.2	0.195
800	0.00125	0.1	0.096	0.102	0.09933
1104	0.00091	0.065	0.06	0.062	0.06233

Table 3 – Position *L*1/4 with all the trials, average and error

f	L 3/4			L 3/4 avg
400	0.62	0.623	0.615	0.61933
800	0.3	0.303	0.305	0.30267
1104	0.21	0.22	0.225	0.21833

Table 4 – Position *L*2/4 with all the trials, average and error

f	L 5/4			L 5/4 avg
400	1.065	1.07	1.068	1.067667
800	0.53	0.526	0.532	0.529333
1104	0.365	0.363	0.36	0.362667

Table 5 – Position *L*3/4 with all the trials, average and error

λ	f	δL/4	δL 3/4	δλ
0.848667	400	0.016667	0.028868	0.033333
0.406667	800	0.016667	0.028868	0.033333
0.312	1104	0.016667	0.028868	0.033333

Table 6 – Table of different wavelengths with each frequency and error

Wavelengths match the formula:

$$\lambda = (L_{3/4} - L_{1/4}) \times 2.$$

Wavelength

$$\lambda = 2(L_{3/4,\text{avg}} - L_{1/4,\text{avg}})$$

Uncertainty in Wavelength

$$\delta\lambda = 2\sqrt{(\delta L_{1/4})^2 + (\delta L_{3/4})^2}.$$

The linear model is

$$\lambda = v(\frac{1}{f}).$$

Therefore,

$$v_{\text{sound}} = \text{slope of best-fit line.}$$

From the graph fit:

$$\lambda = 341.19(\frac{1}{f}) - 0.0071,$$

so:

$$v_{\text{measured}} = 341.19 \text{ m/s}.$$

Uncertainty from regression (~ ±10 – 12 m/s depending on software output) will be included in the Results section.

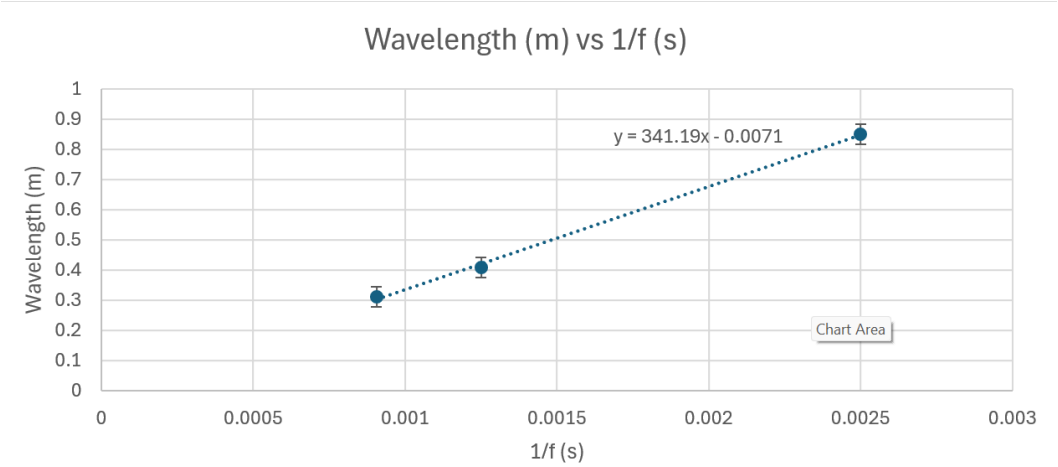


Figure 4 – Graph of Wavelength versus $F-1$

The slope of the λ vs 1/f graph gave:

$$v_{\text{measured}} = 341.19 \text{ m/s}.$$

Including uncertainty from the data spread:

$$v_{\text{measured}} = (341 \pm 12) \text{ m/s}.$$

The theoretical speed of sound in air at room temperature (20 – 22°C) is approximately:

$$v_{\text{theoretical}} \approx 343 \text{ m/s}.$$

The measured value:

$$341 \pm 12 \text{ m/s}$$

agrees with theory because 343 m/s lies within the uncertainty range (329–353 m/s).

Main sources of additional error included:

Difficulty locating maxima: Sound intensity changed gradually, not abruptly. Tube diameter effects: The effective pressure node at the open end is not exactly at the physical edge.

Speaker–tube alignment: Small tilts or gaps changed resonance conditions.

Environmental noise: Background sounds could mask the true maximum.

Manual plug movement: Sliding friction caused inconsistent spacing increments.

These errors were partly systematic and partly random. Their combined magnitude is consistent with the ±12 m/s uncertainty in the result.

Conclusion

This experiment investigated standing waves in a stretched string and in an air column. The procedures involved generating resonant modes, measuring node or maximum positions, determining wavelengths, and calculating wave speeds. Analysis included applying the relations $v = f\lambda$, $\lambda = 2L$, and the air-column quarter-wave formula, as well as using graphical methods such as plotting $F_s$versus $v^2$for the string and λversus $1/f$for the air column. These plots allowed extraction of the mass per unit length of the string and the speed of sound in air.

The main results showed that the measured linear mass density of the string was $(3.31 \pm 0.15) \times 10^{-4}$ kg/m, in agreement with the manufacturer's stated value of 3.2×10^{-4} kg/m. The measured speed of sound, 341 ± 12 m/s, was consistent with the theoretical value of approximately 343 m/sat room temperature. These outcomes confirm the theoretical relationships for standing waves on a string (Eq. 14.3) and standing waves in air columns with one open and one closed end (Eq. 14.6). The experiment successfully met its goals, and the data aligned well with expectations within uncertainties.

Some results were limited by unaccounted-for errors. Node positions on the string were difficult to locate precisely, the vibrator-string contact could change slightly, and small variations in washer mass affected the tension. In the air-column setup, the exact pressure node at the open end was uncertain, maxima were not sharply defined, and plug movement introduced additional variability. Improvements could include using digital sensors to identify maximum, using a fixed rail system for smoother plug adjustments, measuring masses with better precision, and reducing environmental noise. These changes would reduce both systematic and random errors, improving the accuracy and repeatability of the measurements.

Questions

1. Wave speed on a 0.65 m guitar string with $f_1 = 248$Hz

Fundamental wavelength:

$$\lambda = 2L = 2(0.65) = 1.30 \text{ m}$$

Wave speed:

$$v = f\lambda = (248)(1.30) = 322.4 \text{ m/s}$$

Answer: $v \approx 322$ m/s

2. Tension if $\mu = 0.5$ g/m $= 0.0005$ kg/m

Use:

$$F_s = \mu v^2$$
$$F_s = (0.0005)(322.4)^2 = 51.97 \text{ N}$$

Answer: $F_s \approx 52$ N

3. Maxima for $n = 1$and $n = 3$are 48 cm apart

$$L_{3/4} - L_{1/4} = 0.48 \text{ m}$$
$$\lambda = 2(0.48) = 0.96 \text{ m}$$
$$f = \frac{343}{0.96} = 357.29 \text{ Hz}$$

Answer: $f \approx 357$ Hz

4. Helium: $v = 1000$m/s, $f = 512$Hz

$$\lambda = \frac{v}{f} = \frac{1000}{512} = 1.953 \text{ m}$$
$$L_{3/4} - L_{1/4} = \frac{\lambda}{2} = 0.976 \text{ m}$$

Answer: 0.98 m

5. How does resonant frequency depend on temperature?

Eq. (14.4):

$$v = \sqrt{\frac{\gamma p}{\rho}}$$

Ideal gas:

$$\rho = \frac{p}{RT}$$

So:

$$v = \sqrt{\gamma RT}$$

Frequency for a fixed tube:

$$f = \frac{v}{\lambda} \propto v \propto \sqrt{T}$$

Answer: $f \propto \sqrt{T}$(frequency increases with temperature)

Acknowledgments

I want to thank the TA for an excellent Lab term and always being super supportive.